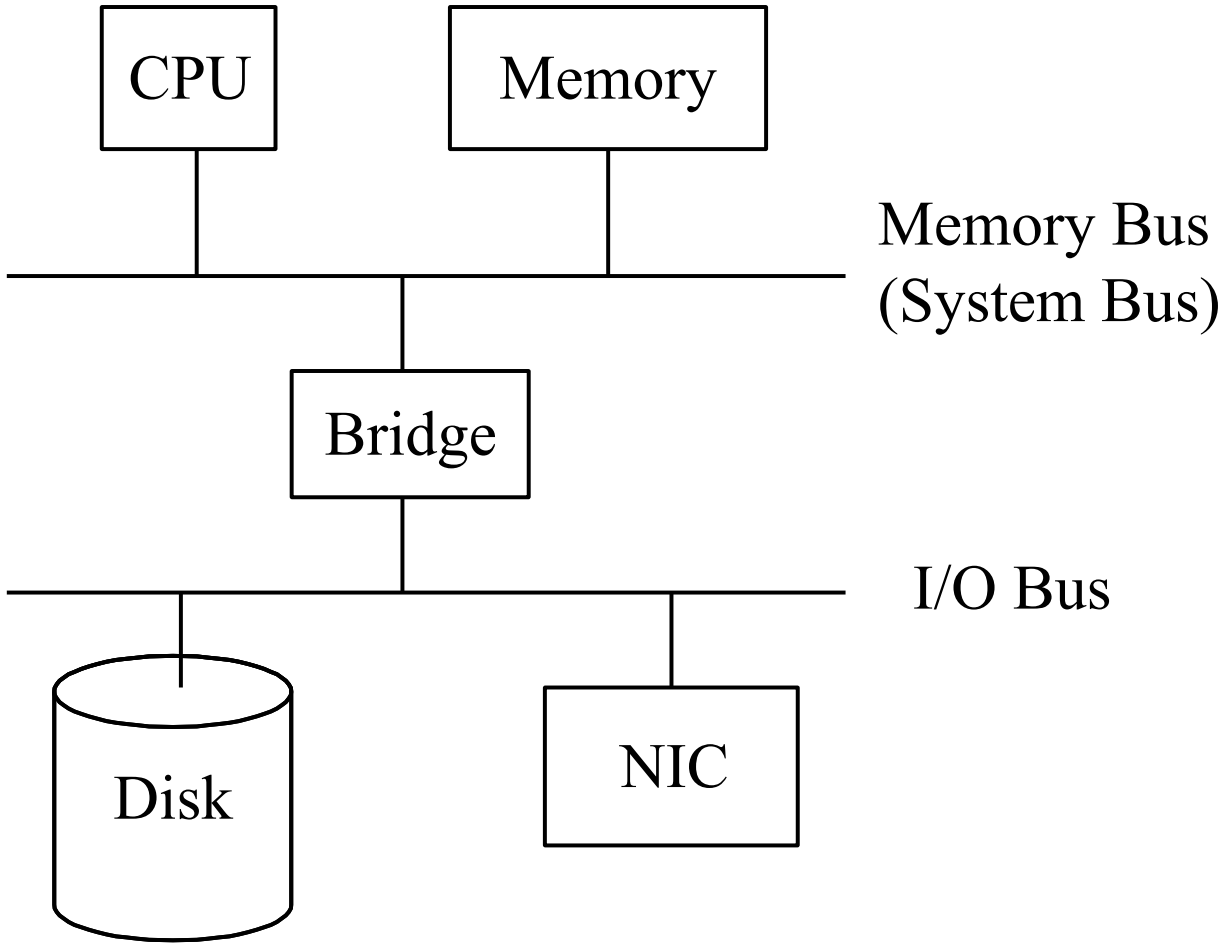
Chapter 12: Disks

CS 416: Operating Systems Design

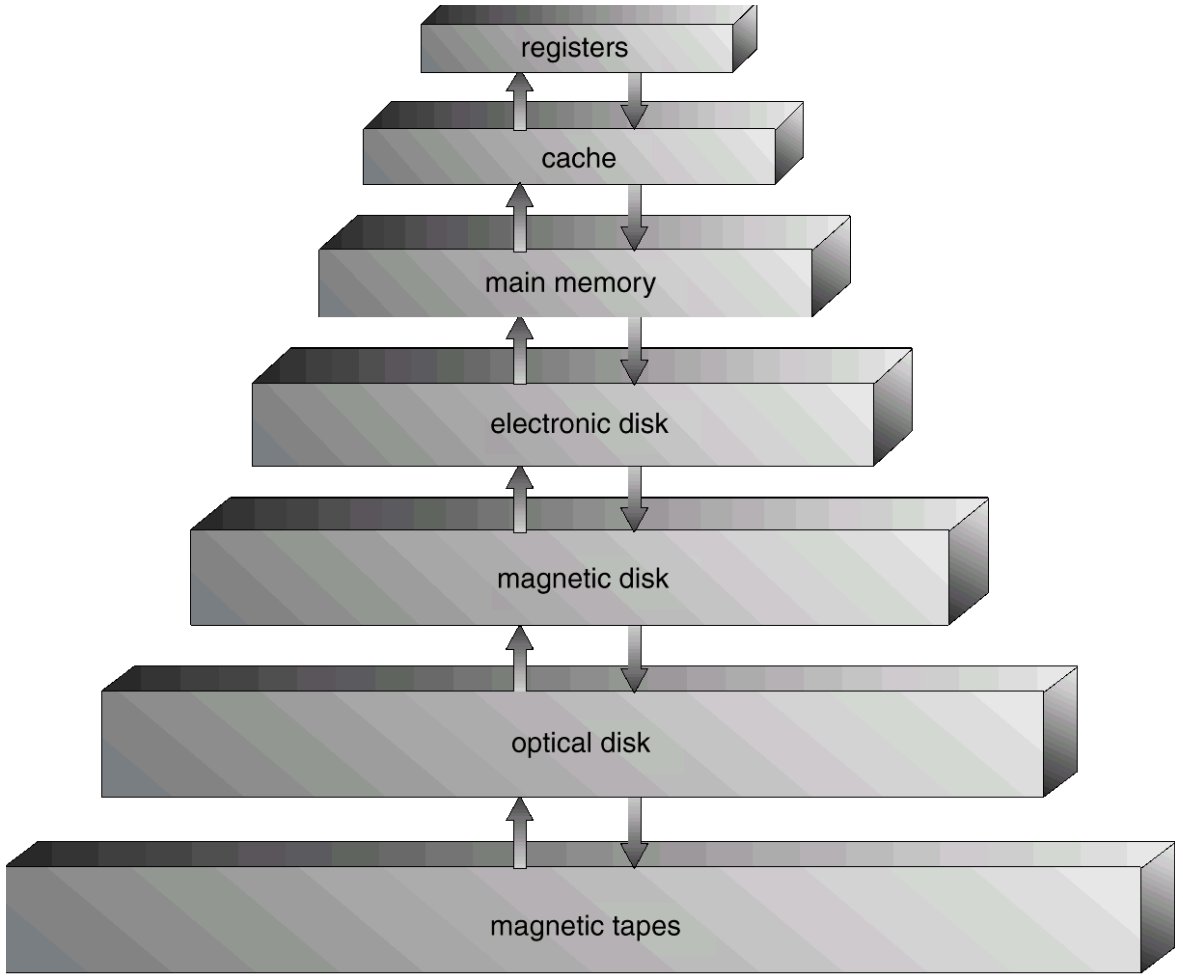
Department of Computer Science
Rutgers University

http://www.cs.rutgers.edu/~vinodg/teaching/416

File System: Abstraction for Secondary Storage



Storage-Device Hierarchy



Secondary Storage

Secondary storage typically:

Is storage outside of memory

Does not permit direct execution of instructions or data retrieval via load/store instructions

Characteristics:

It's large: 100 – 1000 GB (as of March 2008)

It's cheap: 1000 GB IDE disks on order of $300

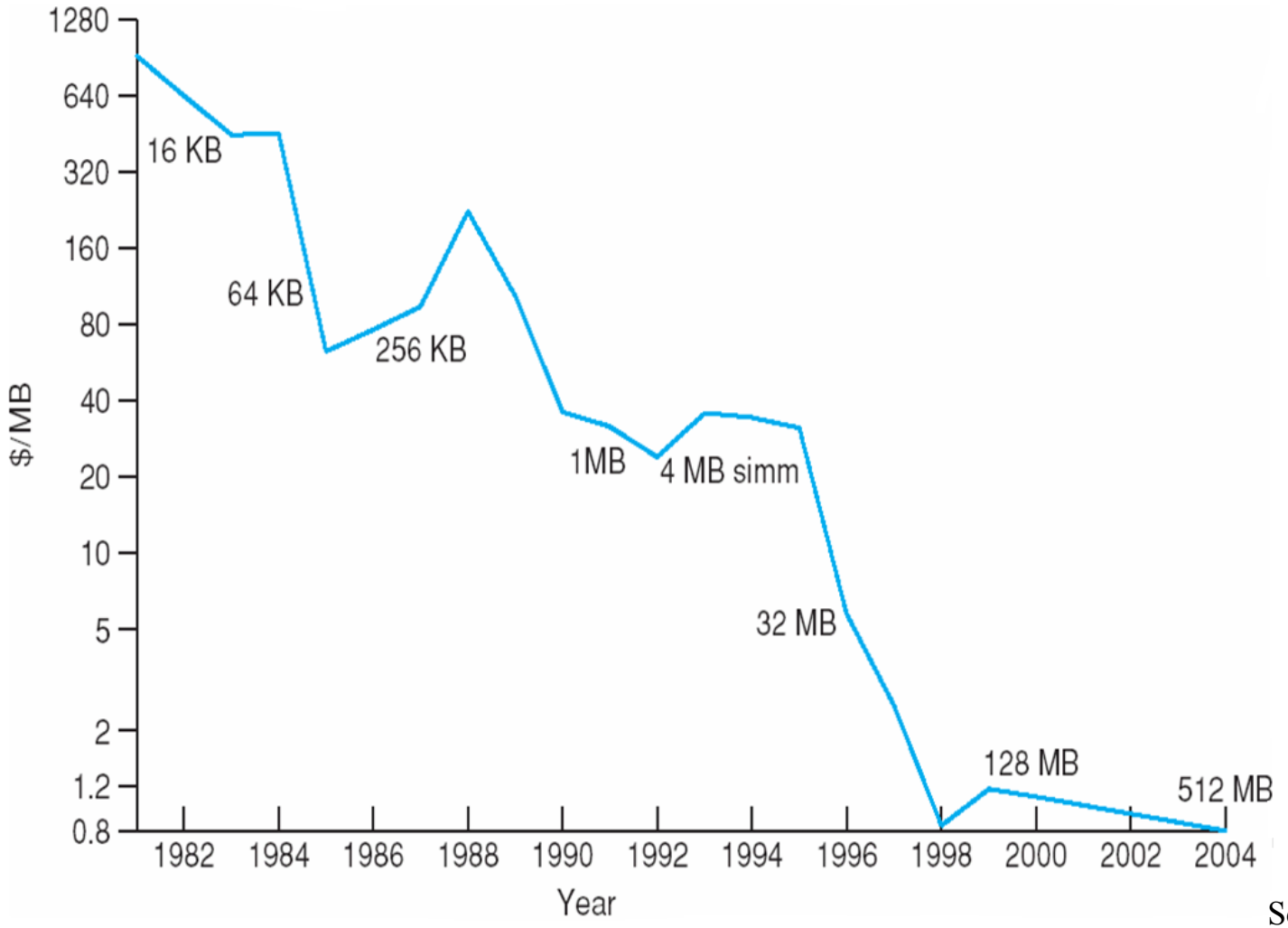
It's persistent: data is maintained across process execution and power down (or loss)

It's slow: milliseconds to access

Costs

- Main memory is much more expensive than disk storage
- The cost/MB of hard disk storage is competitive with magnetic tape if only one tape is used per drive
- The cheapest tape drives and the cheapest disk drives have had about the same storage capacity over the years

Cost of DRAM



Source: SGG

Disk

Magnetic disks provide bulk of secondary storage of modern computers

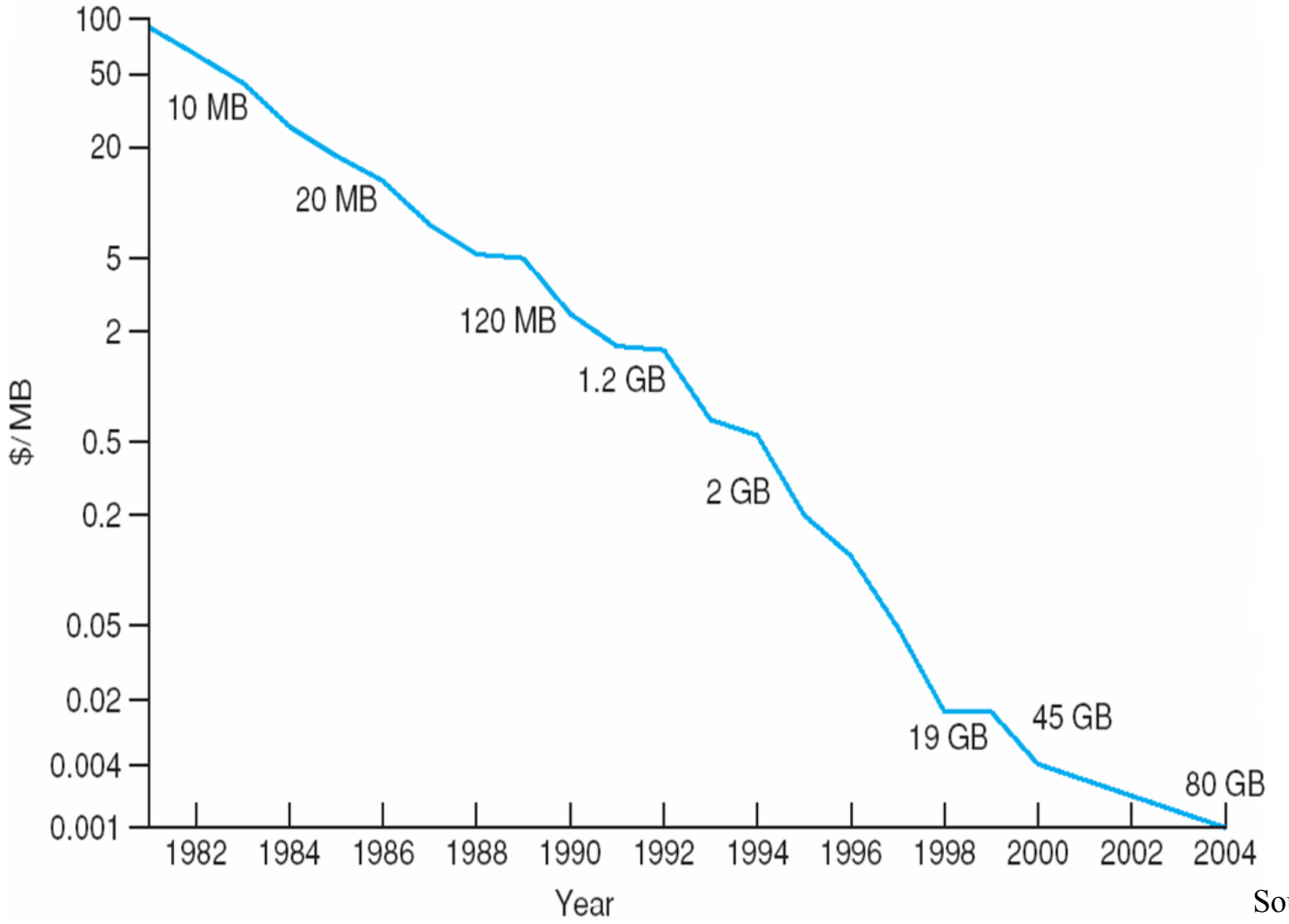
- Drives rotate at 60 to 200 times per second
- **Transfer rate** is rate at which data flow between drive and computer
- **Positioning time** (**random-access time**) is time to move disk arm to desired cylinder (**seek time**) and time for desired sector to rotate under the disk head (**rotational latency**)
- **Head crash** results from disk head making contact with the disk surface
  - That's bad

Disks can be removable

Drive attached to computer via **I/O bus**

- Busses vary, including **EIDE, ATA, SATA, USB, Fibre Channel, SCSI**
- **Host controller** in computer uses bus to talk to **disk controller** built into drive or storage array

Cost of Disks



Source: SGG

Tapes

Magnetic tape

Was early secondary-storage medium

Relatively permanent and holds large quantities of data

Access time slow

Random access ~1000 times slower than disk

Mainly used for backup, storage of infrequently-used data, transfer medium between systems

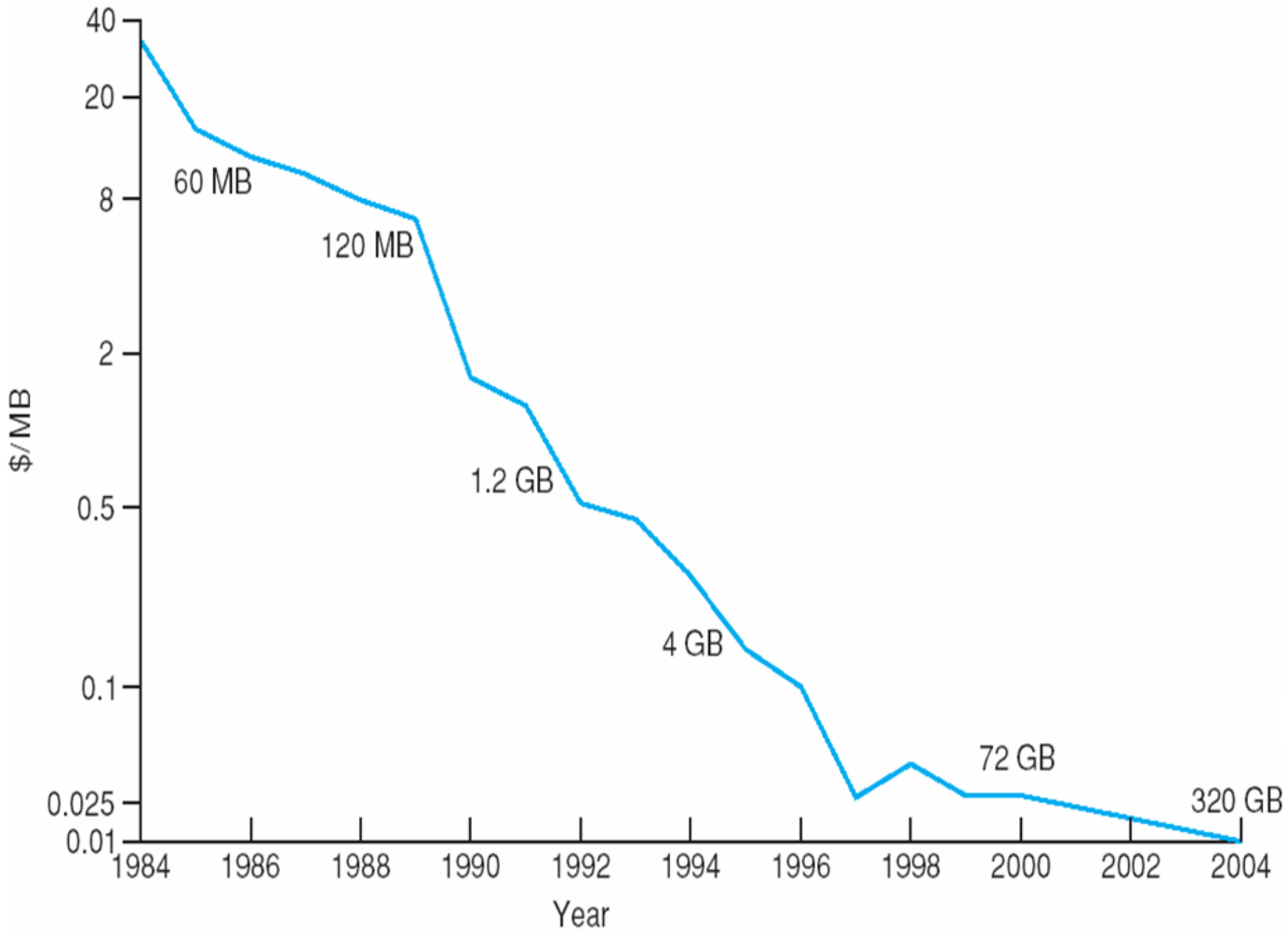
Kept in spool and wound or rewound past read-write head

Once data under head, transfer rates comparable to disk

20-200GB typical storage

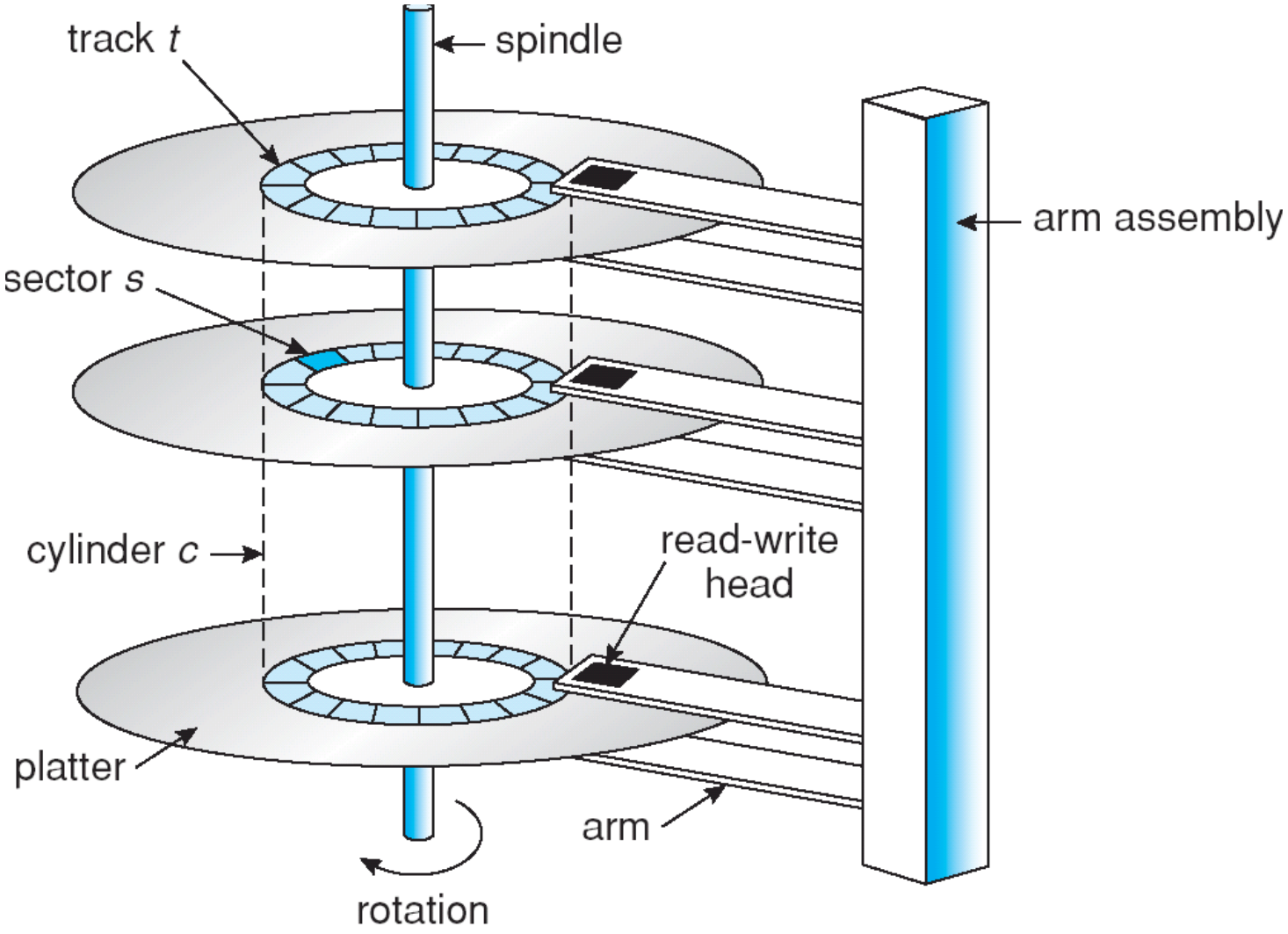
Common technologies are 4mm, 8mm, 19mm, LTO-2 and SDLT

Cost of Tapes

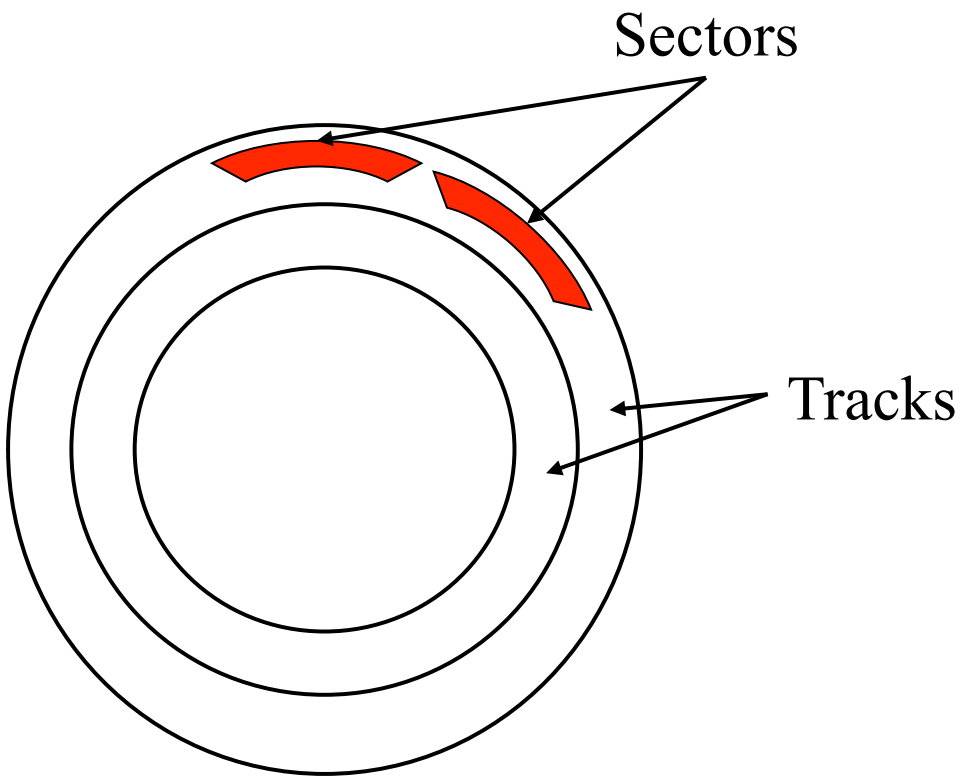


Source: SGG

Moving-head Disk Mechanism



Disks



Seek time: time to move the disk head to the desired track

Rotational delay: time to reach desired sector once head is over the desired track

Transfer rate: rate data read from/written to disk

Some typical parameters:

- Seek: ~5-10ms
- Rotational delay: ~3ms for 10000 rpm
- Transfer rate: 40 MB/s

Disk Structure

Disk drives are addressed as large 1-dimensional arrays of *logical blocks*, where the logical block is the smallest unit of transfer.

The 1-dimensional array of logical blocks is mapped into the sectors of the disk sequentially.

- Sector 0 is the first sector of the first track on the outermost cylinder.
- Mapping proceeds in order through that track, then the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost.

Disk Scheduling

Disks are at least 4 orders of magnitude slower than memory

The performance of disk I/O is vital for the performance of the computer system as a whole

Access time (seek time + rotational delay) >> transfer time for a sector

Therefore the order in which sectors are accessed (read, especially) matters a lot

Disk scheduling

Usually based on the position of the requested sector rather than according to the process priority

Possibly reorder stream of read/write requests to improve performance

Disk Scheduling (Cont.)

Several algorithms exist to schedule the servicing of disk I/O requests.

We illustrate them with a request queue (tracks 0-199).

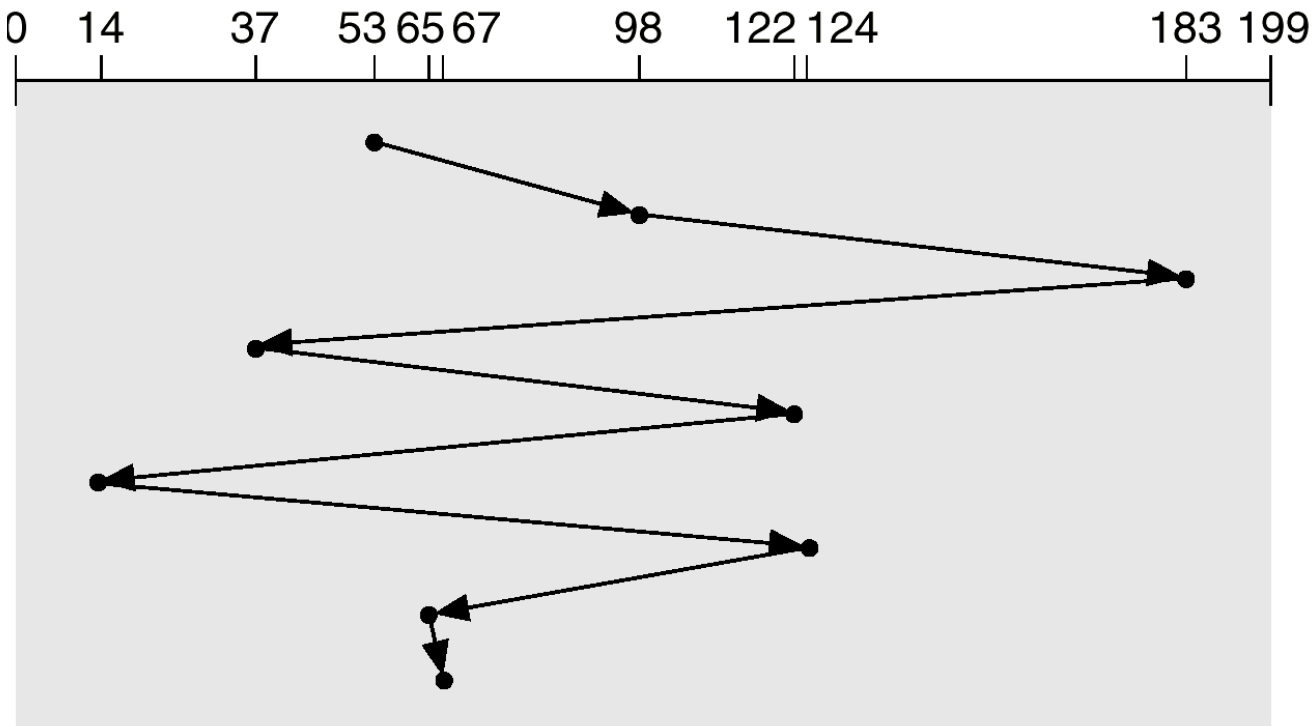
98, 183, 37, 122, 14, 124, 65, 67

Head pointer 53

FCFS

Illustration shows total head movement of 640 cylinders.

queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



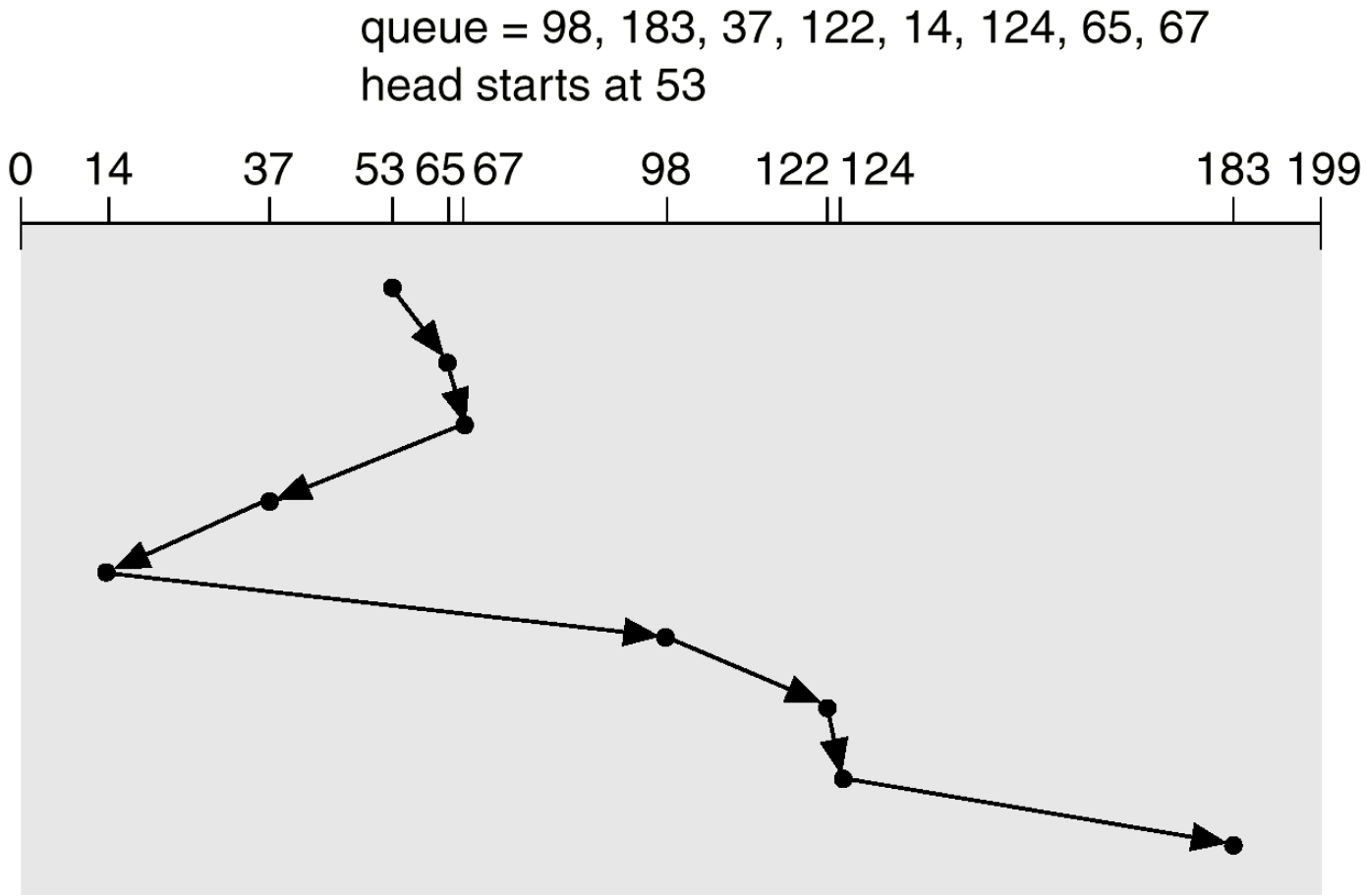
SSTF

Selects the request with the minimum seek time from the current head position.

SSTF scheduling is a form of SJF scheduling; may cause starvation of some requests.

Illustration shows total head movement of 236 cylinders.

SSTF (Cont.)


queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53
0
14
37
53
65
67
98
122
124
183
199

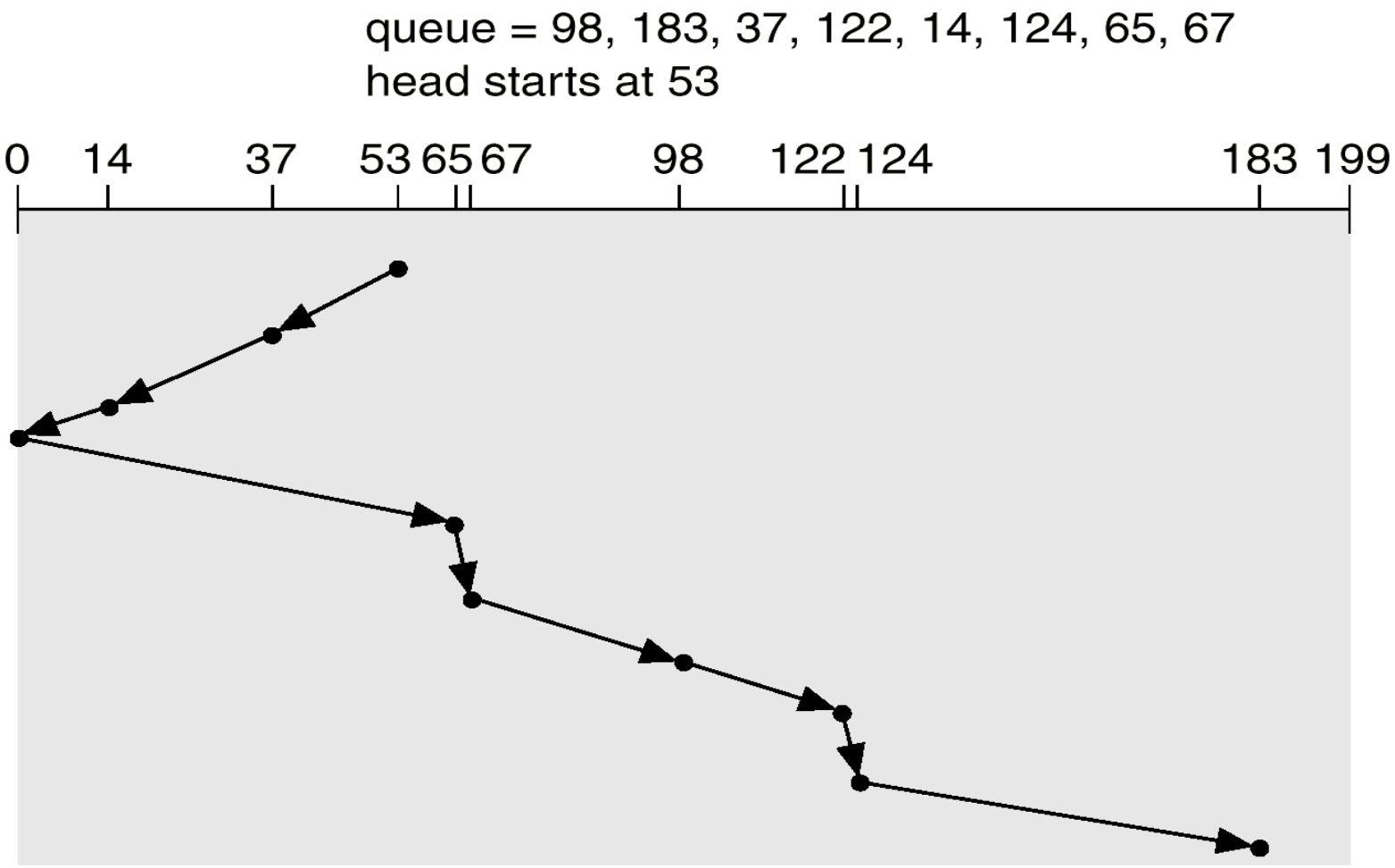
SCAN

The disk arm starts at one end of the disk, and moves toward the other end, servicing requests until it gets to the other end of the disk, where the head movement is reversed and servicing continues.

Sometimes called the *elevator algorithm*.

Illustration shows total head movement of 208 cylinders.

SCAN (Cont.)


queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53
0
14
37
53
65
67
98
122
124
183
199

C-SCAN

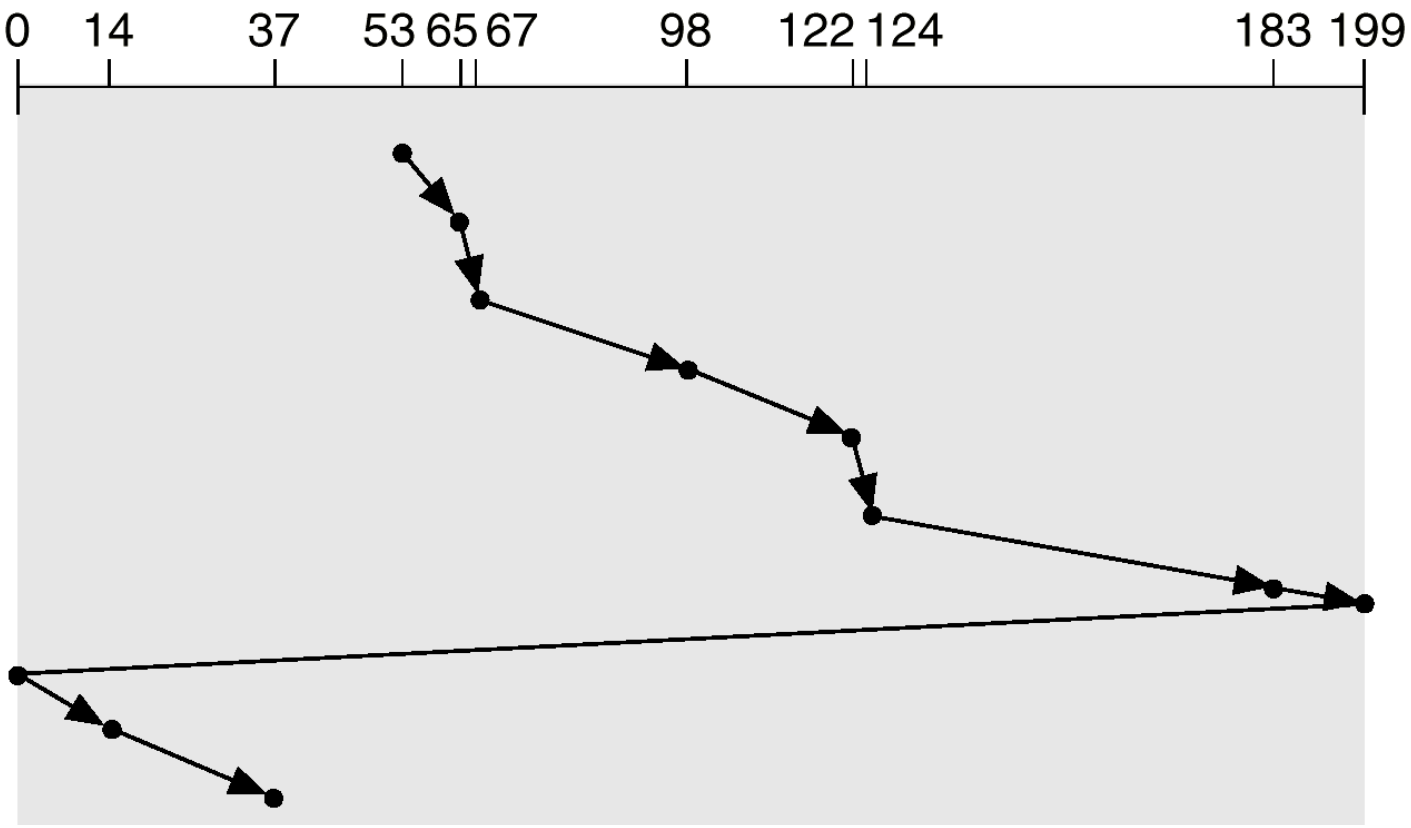
Provides a more uniform wait time than SCAN.

The head moves from one end of the disk to the other, servicing requests as it goes. When it reaches the other end, however, it immediately returns to the beginning of the disk, without servicing any requests on the return trip.

Treats the cylinders as a circular list that wraps around from the last cylinder to the first one.

C-SCAN (Cont.)

queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



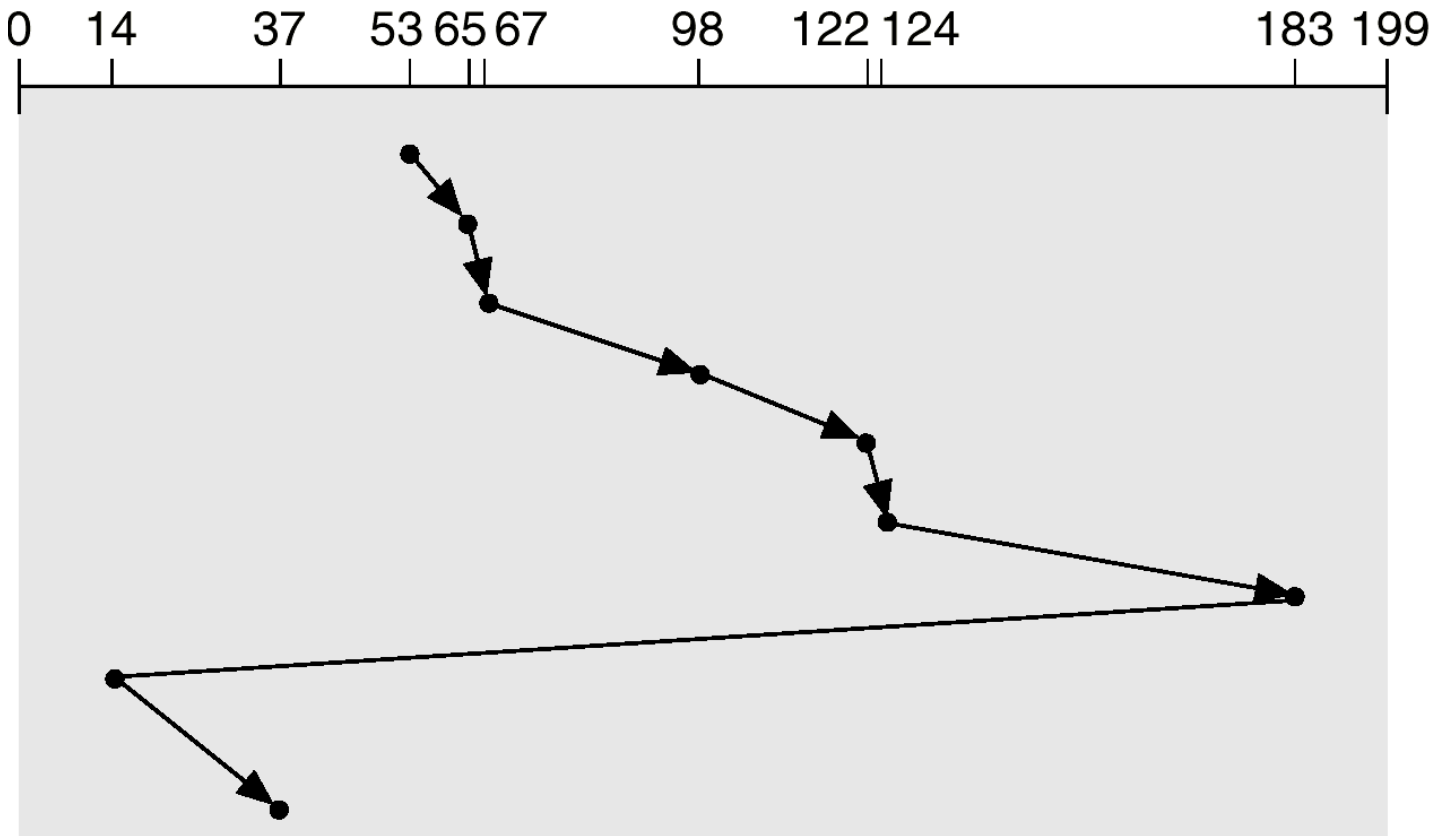
C-LOOK

Version of C-SCAN

Arm only goes as far as the last request in each direction, then reverses direction immediately, without first going all the way to the end of the disk.

C-LOOK (Cont.)

queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



Selecting a Disk-Scheduling Algorithm

SSTF is common and has a natural appeal.

SCAN and C-SCAN perform better for systems that place a heavy load on the disk.

Performance depends on the number and types of requests.

Requests for disk service can be influenced by the file-allocation method.

The disk-scheduling algorithm should be written as a separate module of the operating system, allowing it to be replaced with a different algorithm if necessary.

Either SSTF or LOOK is a reasonable choice for the default algorithm.

Disk Management

Low-level formatting, or *physical formatting* — Dividing a disk into sectors that the disk controller can read and write.

To use a disk to hold files, the operating system still needs to record its own data structures on the disk.

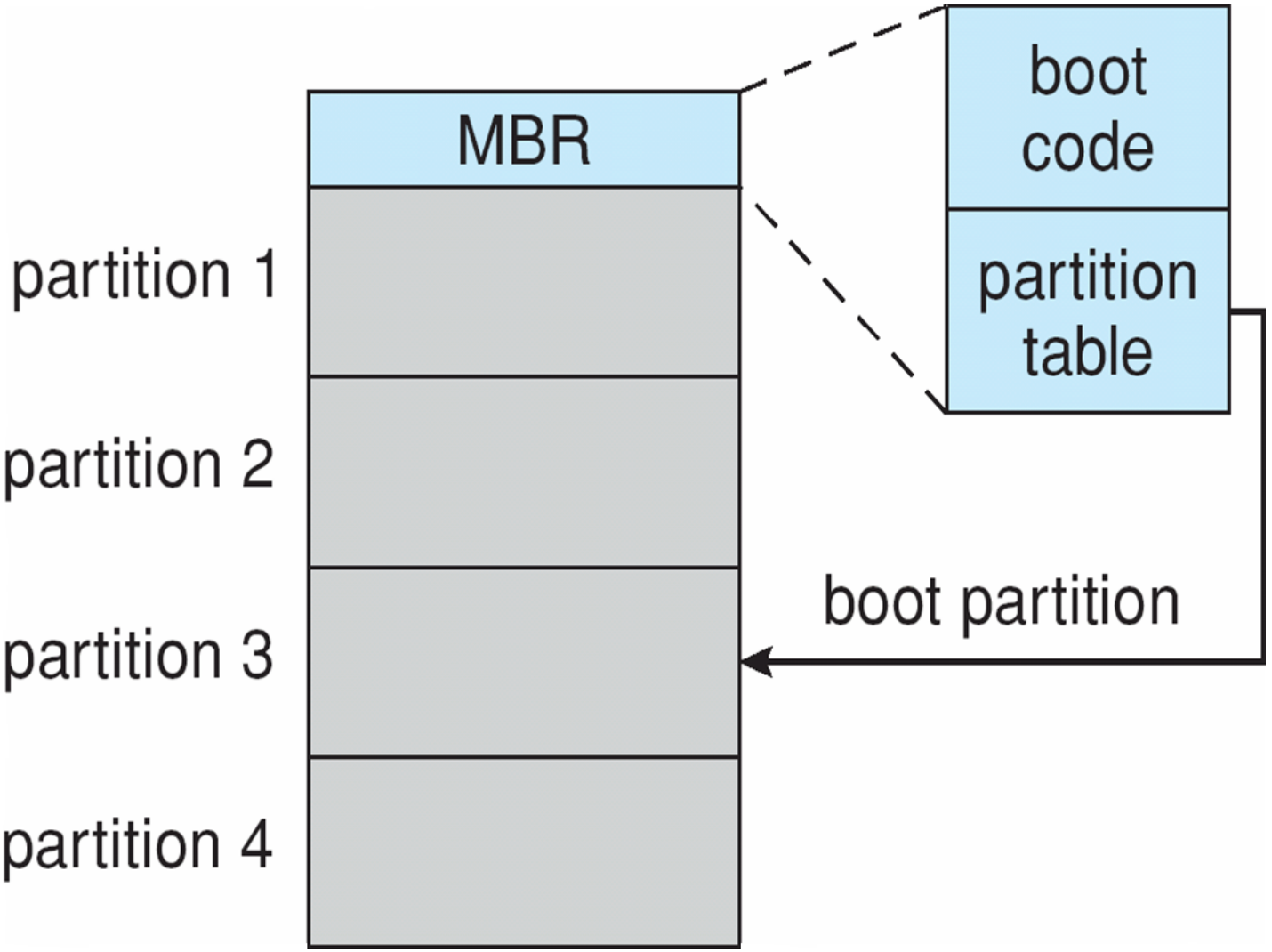
- *Partition* the disk into one or more groups of cylinders.
- *Logical formatting* or "making a file system".

Boot block initializes system.

- The bootstrap is stored in ROM.
- *Bootstrap loader* program.

Methods such as *sector sparing* used to handle bad blocks.

Booting from a Disk in Windows 2000



Swap Space Management

Virtual memory uses disk space as an extension of main memory.

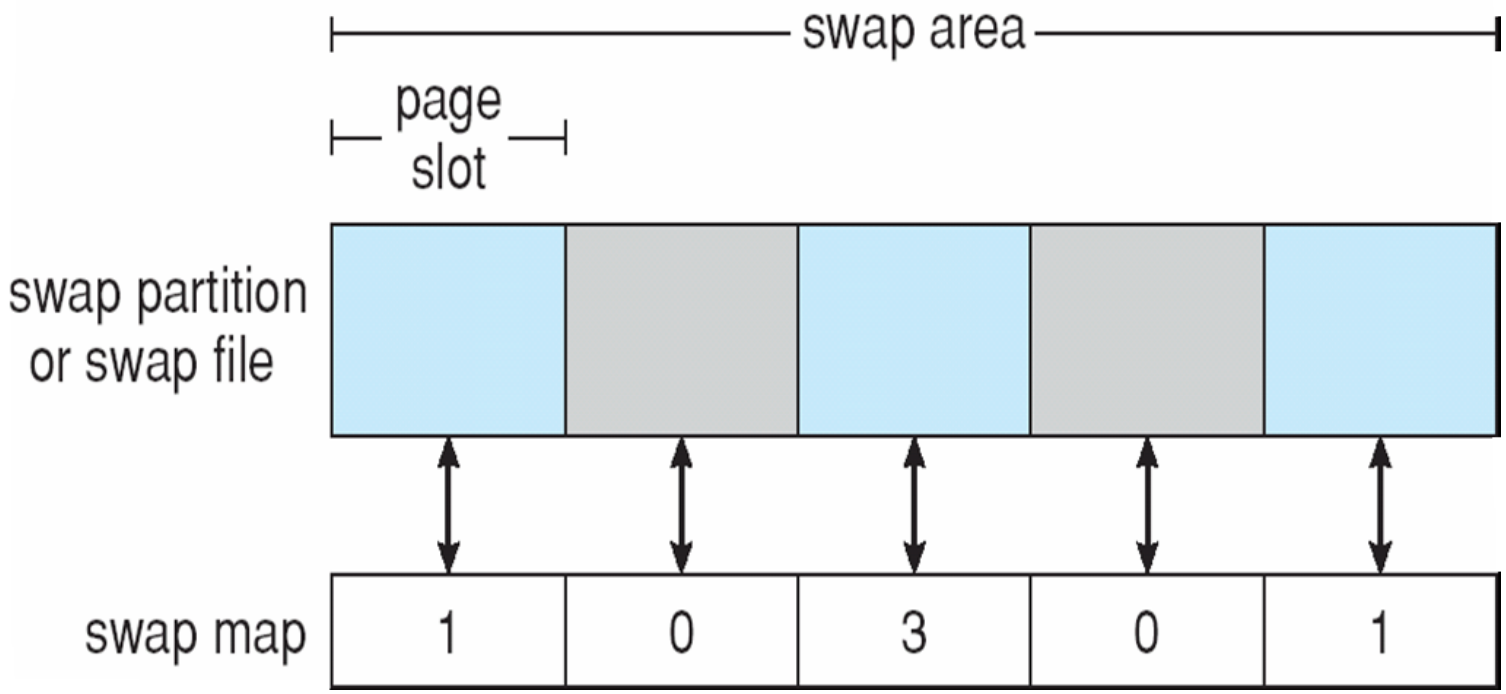
Swap space is necessary for pages that have been written and then replaced from memory.

Swap space can be carved out of the normal file system, or, more commonly, it can be in a separate disk partition.

Swap space management

- 4.3BSD allocates swap space when process starts; swap space holds *text segment* (the program) and *data segment.*
- Kernel uses *swap maps* to track swap-space use.
- Solaris 2 allocates swap space only when a page is forced out of physical memory, not when the virtual memory page is first created.

Data Structures for Swapping on Linux Systems



Disk Reliability

Several improvements in disk-use techniques involve the use of multiple disks working cooperatively.

RAID is one important technique currently in common use.

RAID

Redundant Array of Inexpensive Disks (RAID)

A set of physical disk drives viewed by the OS as a single logical drive

Replace large-capacity disks with multiple smaller-capacity drives to improve the I/O performance (at lower price)

Data are distributed across physical drives in a way that enables simultaneous access to data from multiple drives

Redundant disk capacity is used to compensate for the increase in the probability of failure due to multiple drives

Improve availability because no single point of failure

Six levels of RAID representing different design alternatives

RAID Levels



(a) RAID 0: non-redundant striping.



(b) RAID 1: mirrored disks.



(c) RAID 2: memory-style error-correcting codes.



(d) RAID 3: bit-interleaved parity.



(e) RAID 4: block-interleaved parity.



(f) RAID 5: block-interleaved distributed parity.



(g) RAID 6: P + Q redundancy.

RAID Level 0

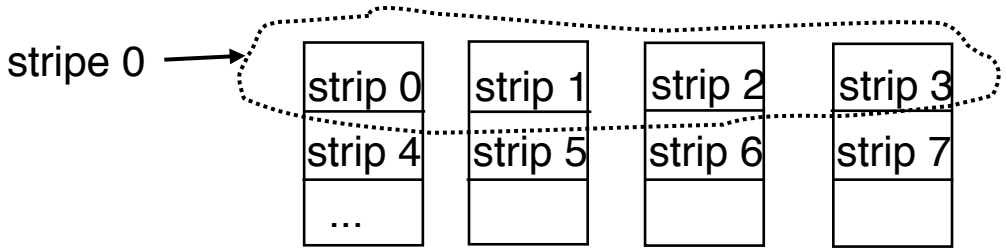
Does not include redundancy

Data are stripped across the available disks

- Total storage space across all disks is divided into strips
- Strips are mapped round-robin to consecutive disks
- A set of consecutive strips that map exactly one strip to each disk in the array is called a stripe

Can you see how this improves the disk I/O bandwidth?

What access pattern gives the best performance?



RAID Level 1

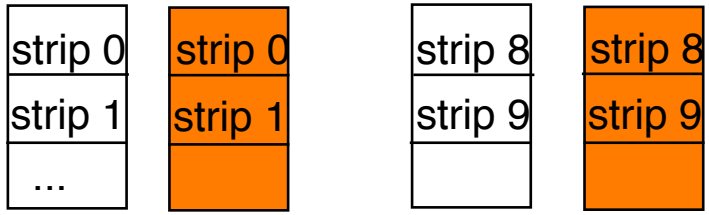
Redundancy achieved by duplicating all the data

Every disk has a mirror disk that stores exactly the same data

A read can be serviced by either of the two disks which contains the requested data (improved performance over RAID 0 if reads dominate)

A write request must be done on both disks but can be done in parallel

Recovery is simple but cost is high



RAID Levels 2 and 3

Parallel access: all disks participate in every I/O request

Small strips (1 bit) since size of each read/write = # of disks * strip size

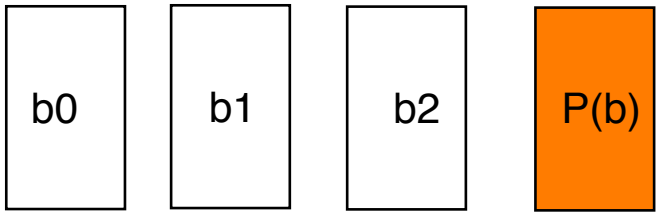
RAID 2: 1-bit strips and error-correcting code. ECC is calculated across corresponding bits on data disks and stored on O(log(# data disks)) ECC disks

- Hamming code: can correct single-bit errors and detect double-bit errors
- Example configurations data disks/ECC disks: 4/3, 10/4, 32/7
- Less expensive than RAID 1 but still high overhead – not needed in most environments

RAID 3: 1-bit strips and a single redundant disk for parity bits

- $P(i) = X2(i) \oplus X1(i) \oplus X0(i)$

On a failure, data can be reconstructed. Only tolerates one failure at a time



$X2(i) = P(i) \oplus X1(i) \oplus X0(i)$

RAID Levels 4 and 5

RAID 4

Large strips with a parity strip like RAID 3

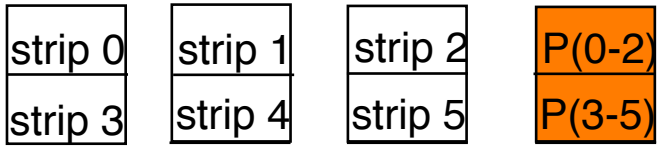
Independent access - each disk operates independently, so multiple I/O request can be satisfied in parallel

Independent access ➜ small write = 2 reads + 2 writes

Example: if write performed only on strip 0:

$$P'(i) = X2(i) \oplus X1(i) \oplus X0'(i)$$
$$= X2(i) \oplus X1(i) \oplus X0'(i) \oplus X0(i) \oplus X0(i)$$
$$= P(i) \oplus X0'(i) \oplus X0(i)$$

Parity disk can become bottleneck

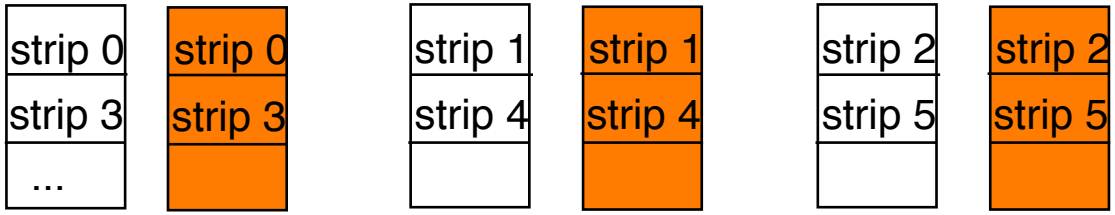


RAID 5

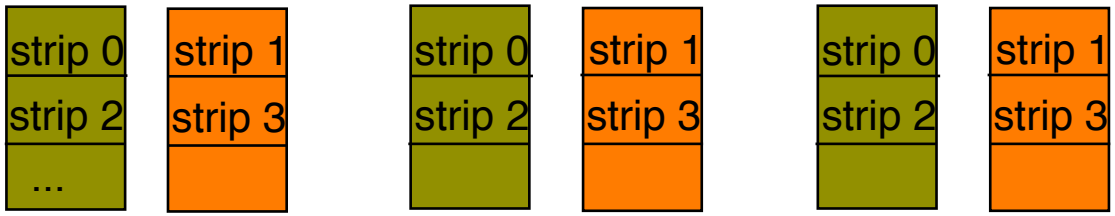
Like RAID 4 but parity strips are distributed across all disks

Other Popular RAID Organizations

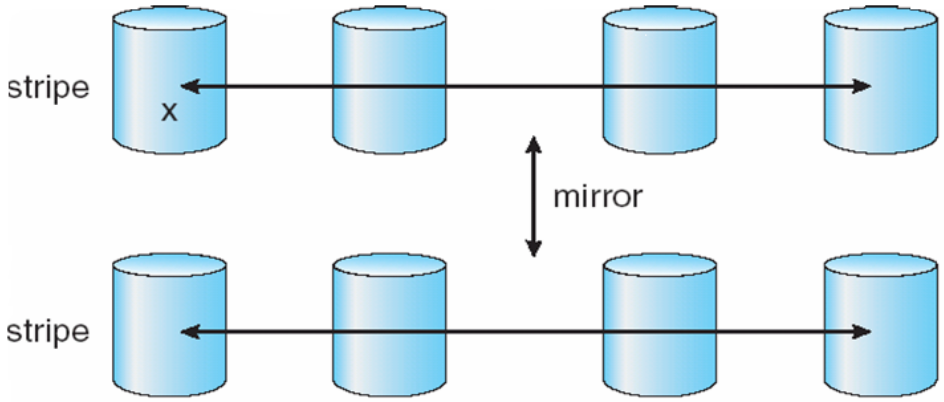
RAID 0 + 1: Stripe first and then mirror



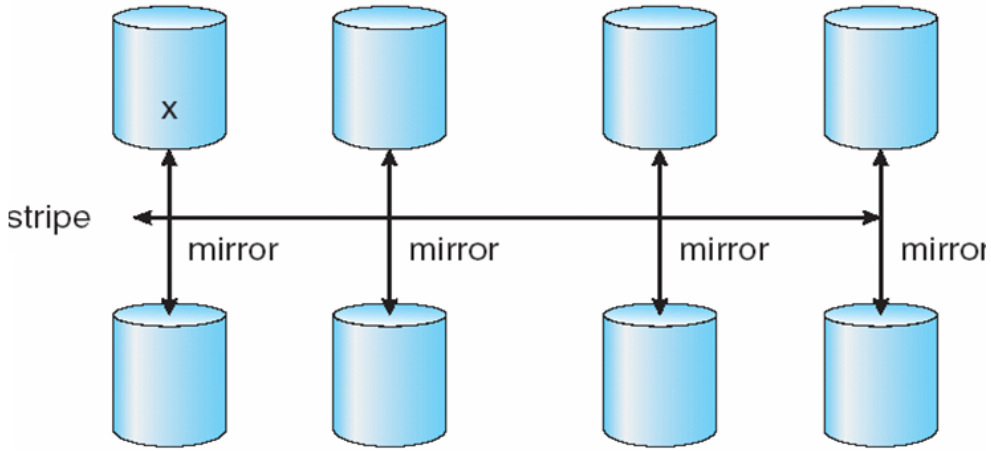
RAID 1 + 0: Mirror first and then stripe



RAID (0 + 1) and (1 + 0)



a) RAID 0 + 1 with a single disk failure.



b) RAID 1 + 0 with a single disk failure.

Disk Attachment

Host-attached storage accessed through I/O ports talking to I/O busses

SCSI itself is a bus, up to 16 devices on one cable, **SCSI initiator** requests operation and **SCSI targets** perform tasks

- Each target can have up to 8 **logical units** (disks attached to device controller

FC is high-speed serial architecture

- Can be switched fabric with 24-bit address space – the basis of **storage area networks** (**SAN**s) in which many hosts attach to many storage units
- Can be **arbitrated loop** (**FC-AL**) of 126 devices

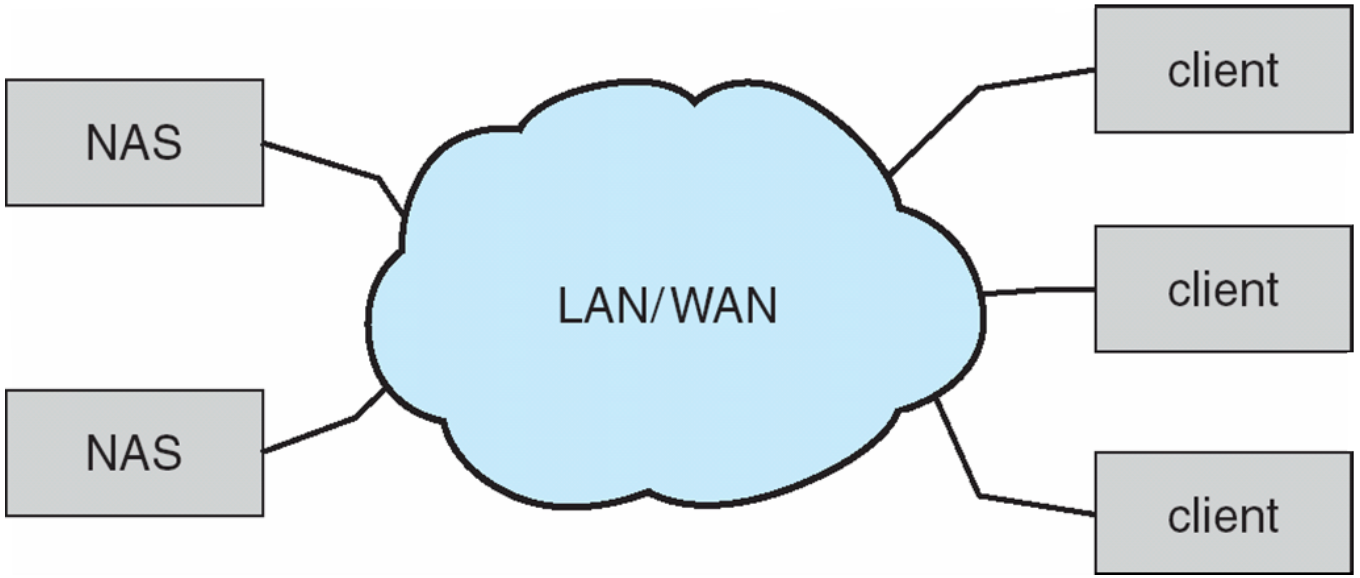
Network-Attached Storage

Network-attached storage (**NAS**) is storage made available over a network rather than over a local connection (such as a bus)

NFS and CIFS are common protocols

Implemented via remote procedure calls (RPCs) between host and storage

New iSCSI protocol uses IP network to carry the SCSI protocol


NAS
NAS
LAN/WAN
client
client
client

Storage Area Network

Common in large storage environments (and becoming more common)

Multiple hosts attached to multiple storage arrays - flexible

